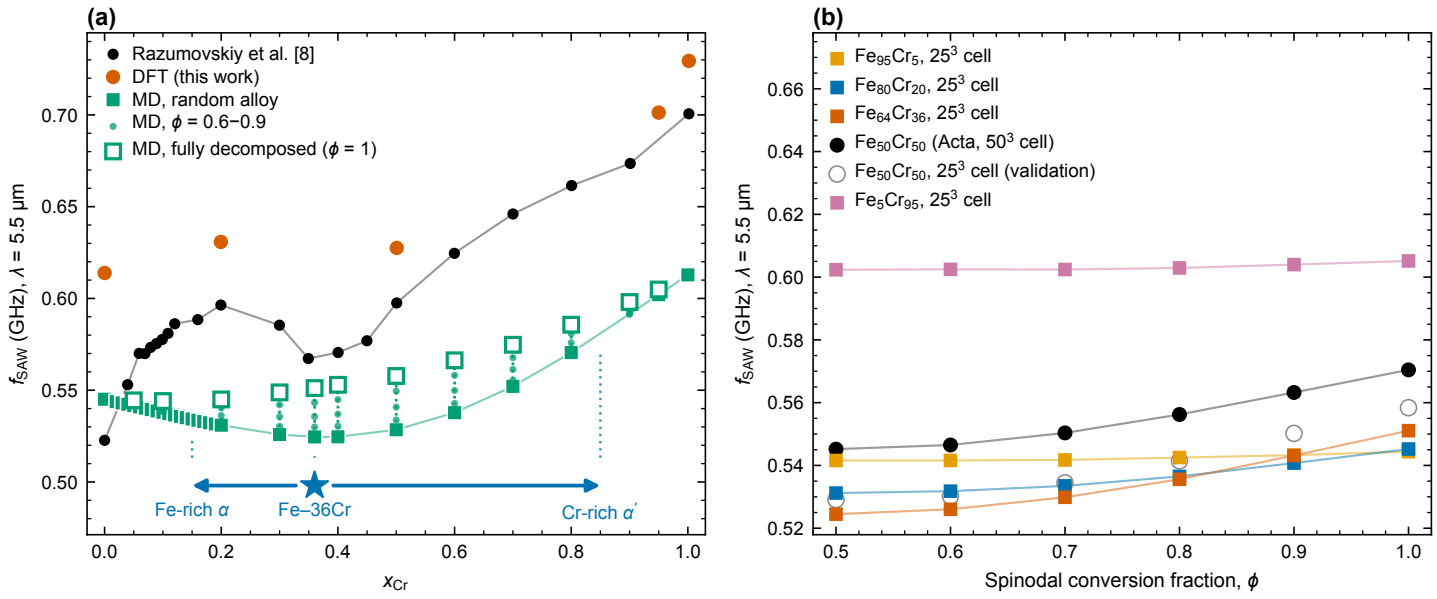

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, $\phi$ = 0.6–0.9
MD, fully decomposed ($\phi$ = 1)
Fe-rich $\alpha$
Fe–36Cr
Cr-rich $\alpha'$
$f_{SAW}$ (GHz), $\lambda$ = 5.5 μm
$x_{Cr}$
0.50
0.55
0.60
0.65
0.70
0.0
0.2
0.4
0.6
0.8
1.0
(b)
$Fe_{95}Cr_5$, $25^3$ cell
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$f_{SAW}$ (GHz), $\lambda$ = 5.5 μm
Spinodal conversion fraction, $\phi$
0.52
0.54
0.56
0.58
0.60
0.62
0.64
0.66
0.5
0.6
0.7
0.8
0.9
1.0